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An EPS Control Strategy to Improve Steering Maneuverability on Slippery Roads

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ABSTRACT

This paper proposes a new Electric Power Steering (EPS) control strategy that improves steering maneuverability especially on slippery roads. In a conventional steering system (including mechanical and hydraulic ones), poor steering wheel returnability associated with reduced alignment torque from the road may lead to awkward handling on slippery roads. In experiments with a test driver, we found that this phenomenon occurs because of the delay in the driver turning the steering wheel to avoid spinning the vehicle. This delay comes from a lower steering wheel returnability than driver expected. Increasing the steering wheel returnability will be effective in avoiding this problem. This can be realized by using the steering angle feedback or the estimated alignment torque feedback. However, the simple feedback of such values will provide drivers with poor road information when the road is slippery. We have developed a new EPS control strategy which certainly provides improved steering wheel returnability and road information on slippery roads. We have tested this control strategy using a test vehicle and confirmed that it provides improved steering maneuverability on slippery roads.

INTRODUCTION

Recently, vehicles equipped with power steering have come into widespread use, and nowadays even micro cars have them. Most power steering systems are hydraulically assisted. However, the fuel consumption of vehicles equipped with hydraulic power-steering is worse than that of mechanical steering systems because the oil pressure must be continuously boosted. EPS systems are on-demand systems that operate only when required by the driver. Hence, using EPS enables us to improve fuel consumption and reduce emissions. This has encouraged the present rapid increase in EPS equipped vehicles.

Conventional power-steering systems operate to reduce the steering torque associated with the alignment torque generated by the tire and road. However, there is no function to augment the steering wheel returnability. Hence, on a slippery road, the reduced alignment torque causes poor steering wheel returnability. When a vehicle begins to spin, drivers turns the steering wheel to avoid spinning. At that time, if the steering wheel returnability is lower than the driver expected, the driver has to apply additional torque to the wheel. This period involves a delay in adjusting the steering and it makes it difficult for the driver to handle the vehicle.

In addition to the steering wheel returnability, road information is also important when driving a vehicle. The driver can perceive that the road is slippery by the change in the steering torque, because of the tires' characteristics.

To realize an EPS which provides both steering wheel returnability and adequate road information, we have developed two new types of control strategy, which improve returnability only when the reaction torque at the steering shaft reaches a predetermined value. The first type is based on the steering angle feedback and the second type is based on the estimated alignment torque feedback.[1]

We have tested this novel control strategy and confirmed its effectiveness in improving steering maneuverability on slippery roads. We have tested the effects under various driving conditions and confirmed that no harmful subjective effects were observed.

STEERING MANEUVERABILITY ON LOW μ ROADS

STEERING MECHANISM - In an EPS system, a motor is applied via a reduction gear to the steering mechanism. This system has both frictional loss torques, and spring characteristics or backlash in the steering mechanism.[2] When a driver steers, alignment torque is generated by the tires and the road surface.

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The operation of EPS can be explained by Figure 1. When the driver steers, the steering torque is detected by a torque sensor placed between the steering wheel and motor. According to the torque sensor signal, the electric motor current is determined. The torque of the electric motor is input to the steering shaft through a reduction gear and the product of the motor torque and the gear ratio act as torque assistance for the driver. The sum of the driver's steering torque and this assisting torque rotates the steering shaft against the alignment torque from the tires and the frictional losses in the steering mechanism. To control the EPS system, the supply voltage and current to the motor are also detected in addition to the steering torque.

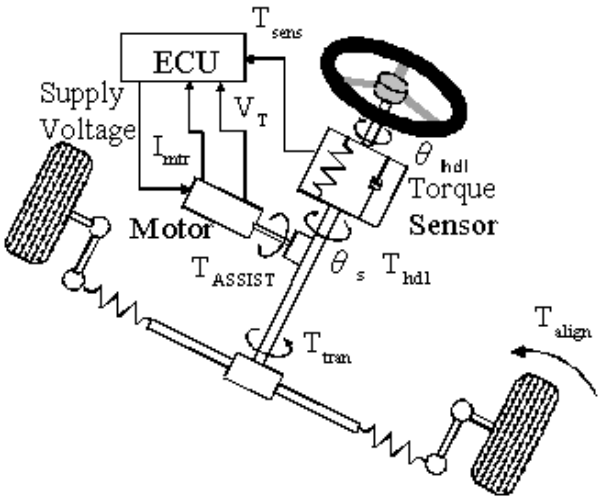


Figure 1 Steering mechanism with EPS

REACTION TORQUE AT THE STEERING SHAFT – The tire characteristics on a low μ road are shown in Figure 2. As can be observed, linear regions exist in both of the side force and the alignment torque. On a high μ road, these linear regions are large and the slip angle of the tires seldom deviates from these regions. However, on a low μ road the linear regions are very small and they occasionally deviate from these regions. For on-center handling, the driver usually steers within the side force and alignment torque linear region, hence the steering maneuverability is almost the same as that on a high μ road. However, the alignment torque linear region is smaller than that of the side force. As the alignment torque linear region is almost the same as the on-center handling regions, the slip angle of the tires occasionally deviates from the linear region when it is located within the side force linear region. When the driver perceive that the slip angle of the tire deviates in the alignment torque linear regions, the driver will begin to make a steering adjustment, to be specific the driver will turn the steering wheel to prevent the vehicle spinning. A driver handles a vehicle on a low μ road In this manner.

Now consider the motion of the a driver turning the steering wheel back. When the driver turns the steering wheel back, the driver usually loosens his grip on the steering wheel in expectation of the self-centering effect. If the alignment torque is smaller than the driver's expectations, the steering wheel may stick or returns very slowly. To turn the steering wheel, the driver will apply extra torque. However the delay of this extra action will affect the steering maneuverability on a low μ road.

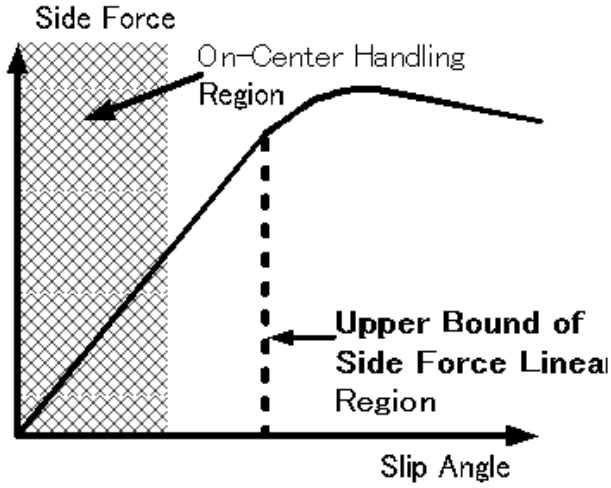


Figure 2a Slip angle vs. side force on a slippery road

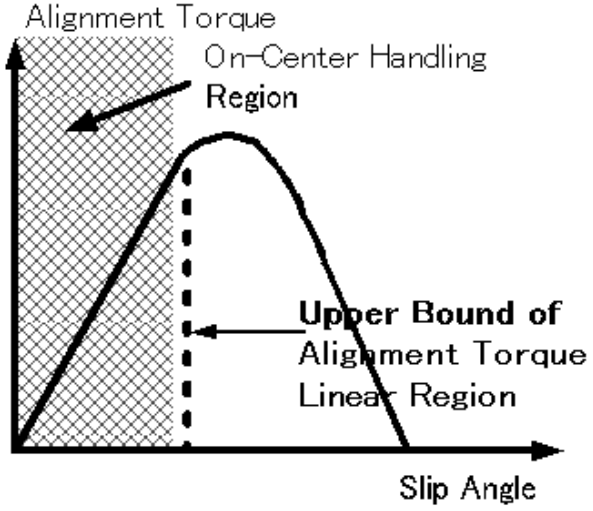


Figure 2b Slip angle vs. alignment torque on a slippery Road

STEERING MANEUVERABILITY – As has been explained, the key to steering maneuverability on a low μ road are as follows.

(1) steering wheel returnability

(2) slippery road information conveyed to the driver through steering torque

A new controller has therefore to be developed to meet these specifications.

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CONTROL STRATEGY

CONCEPT OF NEW METHOD - We explained the problems and the key to steering maneuverability on low μ roads in the preceding section. In this section, we give new control strategy for EPS. This algorithm was developed using the following concepts.

(1) Improved steering wheel returnability on low μ roads as on high μ roads.
(2) Not to lose the sense of slipperiness, on low μ roads.
(3) Freedom from harmful subjective effects even on high μ roads.

OUTLINE OF CONTROL STRATEGY - We aim to improve steering maneuverability and steering wheel returnability on low μ roads by using a new control algorithm called "Steering Angle Feedback Control". A block diagram of this controller is shown in Figure 3. As we can see, first in this method we estimate the reaction torque at the steering shaft (RTSS) (this is composed of alignment torque and friction torque) at "RTSS Observer". Next, according to the estimated RTSS, the steering angle feedback gain is automatically varied.

The variable gain is set so that the sense of slipperiness is not lost. This is described in the "Steering Angle Feedback Gain" section. As shown in Figure 3, it is not necessary to estimate the μ of the road in this control strategy.

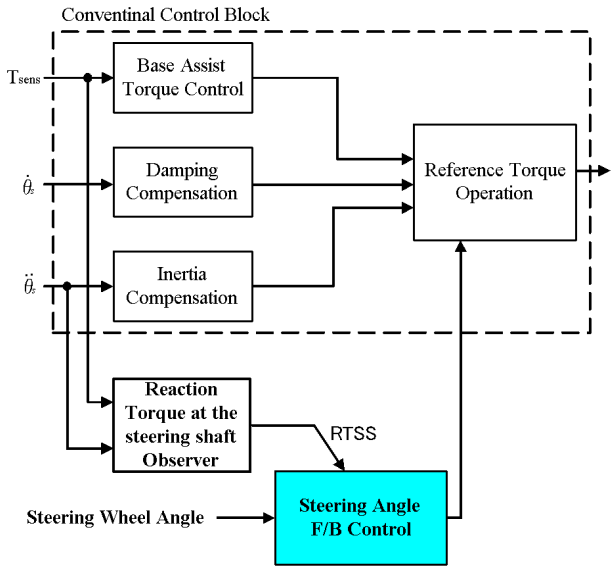


Figure 3 Block diagram of the proposed controller

RTSS OBSERVER - The steering mechanism is expressed by Equation 1.[1] Here,"Ttran" denotes the reaction torque at the steering shaft. This is RTSS itself. This is the sum of alignment torque from the tires "Talign" and the friction loss torque of the rack & pinion etc "Tfric".

$$J\ddot{\theta}=T_{ASSIST}+T_{hdl}-T_{tran} \quad (1a)$$

where

$$T_{ASSIST}=G_{gear}\cdot\left\{K_T\cdot I_{mtr}-T_{f_mtr}\cdot\mathrm{sgn}\left(\dot{\theta}_S\right)\right\}$$
$$T_{tran}=T_{align}+T_{f_rp}\cdot\mathrm{sgn}\left(\dot{\theta}_S\right) \quad (1b)$$

Equations 1a and 1b yield Equation 2.

$$T_{Align}+\left(G_{gear}\cdot T_{f_mtr}\cdot T_{f_rp}\right)\cdot\mathrm{sgn}\left(\dot{\theta}_S\right)=T_{hdl}+G_{gear}\cdot K_T\cdot I_{mtr}-J\ddot{\theta}_S \quad (2)$$

The RTSS observer estimates the steering reaction torque at the steering shaft, which is composed of the alignment torque and friction torque in the steering mechanism.

STEERING ANGLE FEEDBACK GAIN – The steering wheel angle feedback gain is set as shown in Figure 4. It is set to be high when RTSS is small, and low when RTSS is large.

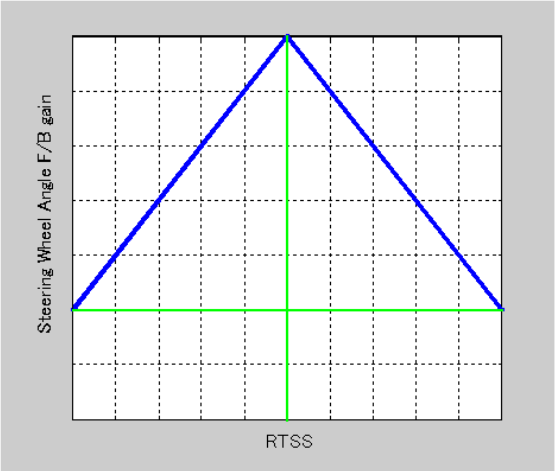


Figure 4 RTSS vs. steering angle feedback gain

Even if the steering wheel angle is same, the RTSS changes depending on the direction of steering. The relationship between the estimated RTSS and the steering angle feedback torque with this method is illustrated in Figure 5. The RTSS is larger when the driver turns the wheel to increase the steering wheel angle (turn), because the direction of the friction torque is in the same direction as the alignment torque. On the other hand, it is smaller when the driver turns the wheel back to decrease the steering wheel angle (return), because the direction of friction torque is in the opposite direction to the alignment torque. Therefore, steering angle feedback gain will be a lower value when the driver turns the steering wheel and it will be a higher value when driver turns the steering wheel back. As a result, if the driver's turn exceeds the alignment torque liner region, the steering angle F/B gain will be about zero (A of Figure 5), because the alignment torque is still large unless the driver turns the steering wheel too much. Therefore, the driver feels the sense of slipperiness

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on a low μ road the same as conventional control which satisfies Concept 2: Not to lose the sense of slipperiness. When the driver begins to turn the steering wheel back the steering angle feedback gain will be larger, because the direction of the friction torque changes, and RTSS is small. As a logical consequence, the steering wheel angle feedback torque becomes larger (B of Figure 5). Therefore, the driver feels the torque generated by this algorithm and perceives on-center feeling.

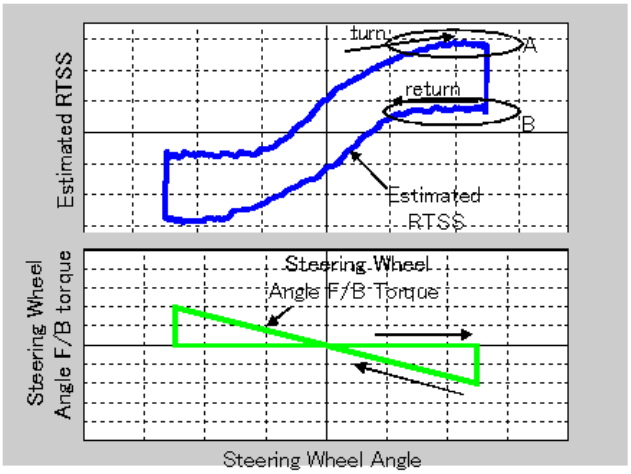


Figure 5 Steering wheel angle vs. "RTSS" and "steering angle feedback torque"

FURTHER DEVELOPMENT OF SENSORLESS CONTROLLER – We also developed a control strategy which does not require steering wheel angle sensor. In this control strategy, we used the estimated alignment torque in instead of the measured steering wheel angle. This algorithm will enable a reduction of costs because a sensor is not required. The effectiveness of this sensorless algorithm is shown by the following vehicle test result.

SIMULATION

SIMULATION CONDITIONS - In this section, we examine the new method by simulation. The simulation conditions are as follows. The steering pattern is sinusoidal steering which emulates the vehicle changing lane.[3]

Table 1 Simulation conditions

Steering Pattern	Sinusoidal steering
Road Condition	μ = 0.2 (such as an icy road)
Velocity (Kph)	40
Control specification	• Conventional Strategy • Proposed Control Strategy

SIMULATION RESULT –The results of simulation of the steering 'feeling' is shown in Figure 6. In the figure, the upper row on the left side denotes the steering wheel angle and the lower row on the left side denotes the steering torque. On the right side, the x-axis denotes the steering wheel angle and the y-axis denotes the steering torque. When the driver begins to turn the steering wheel back, the steering torque of the proposed controller is larger than that of a conventional controller. This will certainly help the driver to turn the steering wheel back. Also, a reduction of the steering torque is observed when the driver's steering exceeds the alignment torque linear region. This reduction of the steering torque will be perceived by the driver as a slippery road.

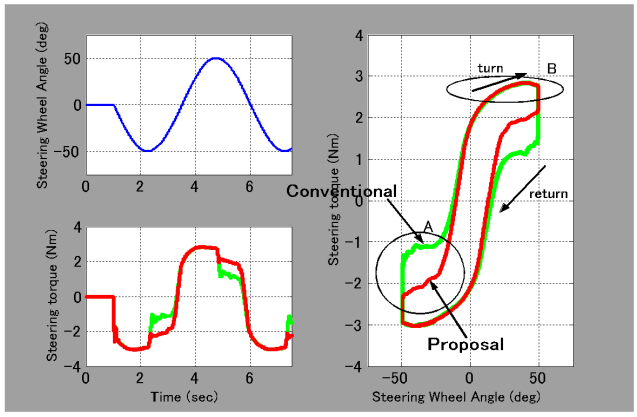


Figure 6 Simulation results

EXEPERIMENT

TEST CONDITIONS – We tested the proposed control strategy using a test vehicle. The test vehicle was a compact car with a 1200cc SI engine. In this experiment, the steering patterns we selected were the release of the steering wheel, sinusoidal steering and a slalom. All of them were performed at 40Kph.

Release of steering wheel – The result of testing the release of the steering wheel using a conventional controller, the steering angle feedback controller, and the estimated alignment torque feedback controller are shown in Figure 7. In this figure, the steering wheel angle is shown, and steering wheel was released from 100 degrees. In testing on a low μ road, the slip angle corresponds to the region where it considerably exceeds the alignment torque linear region. From the results, we see that on a low μ road, for steering angle feedback and estimated alignment torque feedback, the steering wheel returns to center faster than when using the conventional strategy.

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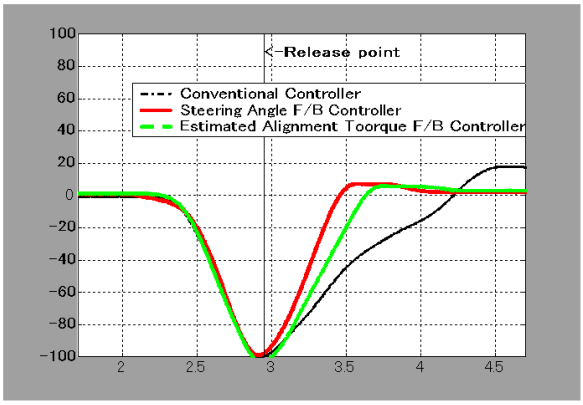


Figure 7 Test of releasing the steering wheel on a low μ road

Sinusoidal steering - The result of testing the sinusoidal steering are shown in Figure8 and 9. In Figure 8, the upper row on the left side denotes the steering wheel angle and the lower row on the left side denotes steering torque. On the right side, the x-axis denotes the steering wheel angle and the y-axis denotes the steering torque. Figure 8 shows the results with the conventional controller and an enlarged view of part of Figure 8 is shown in Figure 9. Figure 9 shows the results with the conventional controller, the steering angle feedback controller, and the estimated alignment torque feedback controller on a low μ road. The following conclusion can be drawn.

- On a low μ road, using the steering angle feedback and the estimated alignment torque feedback, the driver feels increased torque when he turns the steering wheel back. In contrast, when using the conventional control strategy, the driver feels a reduced torque when he turns the steering wheel back. (A in Figure 9) As was mentioned above, the increased returning torque certainly improves steering maneuverability
- On low μ roads, the proposed controllers give the driver the same sense of slipperiness as the conventional controller. (B in Figure 9)

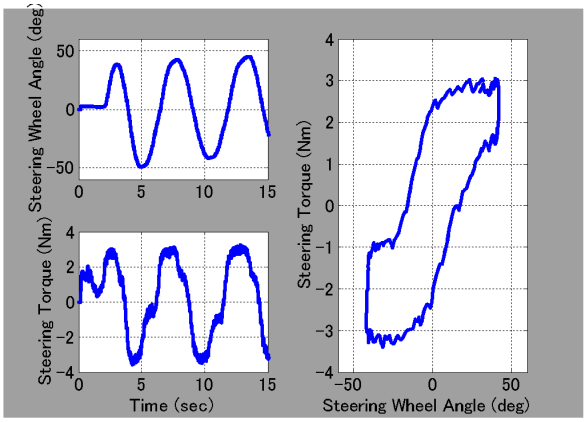


Figure 8 Test of sinusoidal steering with conventional controller on a low μ road

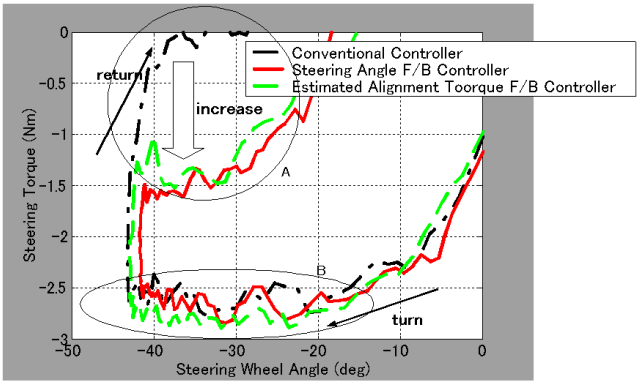


Figure 9 Vehicle test of sinusoidal steering with each controller on a low μ road

Slalom test - The results of the slalom test are shown in Figure 10. This shows the maneuverability of the test vehicle on a low μ road. In Figure10a and 10b, the upper row denotes the steering wheel angle and the steering rate, the middle row denotes the steering torque, and the lower row denotes the yaw rate. Figure 10a shows the results with the conventional controller. Figure 10b shows the results with the steering angle feedback controller. As a performance index, we calculated the average torque 0.4 seconds after the driver began to turn the steering wheel back. The larger the average torque, the easier the turning the steering wheel back becomes. This index is shown in Table 2. In Figure 10, the road condition is a low μ road. The following conclusions can be drawn from the results.

- The difference in the steering rate of returning can be observed mainly when the driver begins to turn the steering wheel back. As can be seen from the steering rate signal, it takes about 0.5 seconds to change the steering wheel from the maximum rate to the maximum rate in the opposite direction when using the proposed controller, while it takes about 1.9 seconds using the conventional controller.
- In the conventional strategy, the returning torque is small because alignment torque is small on a low μ road. (Table 2) This leads to delay in returning the steering wheel. As a result, the yaw rate of the vehicle increases after driver returns the steering wheel. (A in Figure 10a) It is considered that this is harmful for drivers because of the worsened maneuverability.

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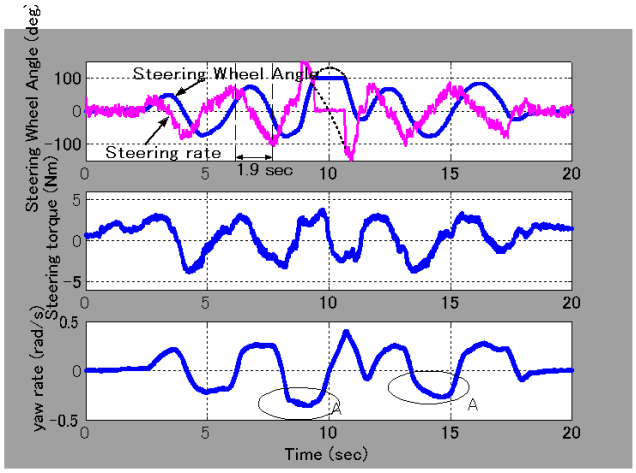


Figure 10a Test of slalom handling with the conventional controller on a low μ road

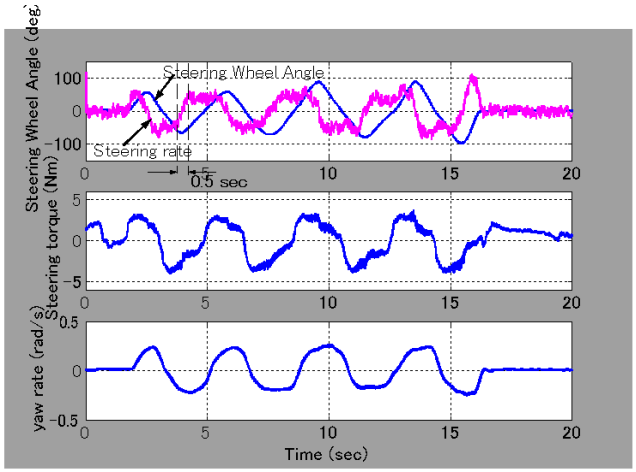


Figure 10b Test of slalom handling with the steering angle feedback controller on a low μ road

Table2 Average torque 0.4 seconds after the driver begins to return the steering wheel

Controller	**Average torque (Nm)**	Maximum average torque (Nm)	Minimum average torque (Nm)
Convention	**0.32**	0.81	-0.4
Steering angle F/B	**1.78**	1.90	1.65
Estimated alignment torque F/B	**1.14**	1.41	0.78

Test results on a high μ road - The results of testing on a high μ road are shown in Figure11 and 12. In these figures, the steering wheel angle is shown. Figure 11 shows the result of releasing the steering with the conventional controller, the steering angle feedback controller and the estimated alignment torque feedback controller. Figure 12 shows the result of sinusoidal steering. The x-axis denotes the steering wheel angle and the y-axis denotes the steering torque. The following conclusions can be drawn from the results.

- Using the proposed strategies, steering wheel returns to center as fast as conventional controllers (Figure 11).
- Using the proposed strategies, the on-center feeling is improved because the steering wheel angle at 0 Nm, which corresponds to returnability, improves (Figure 12).

Hence, this proposed control strategy is free from harmful subjective effects.

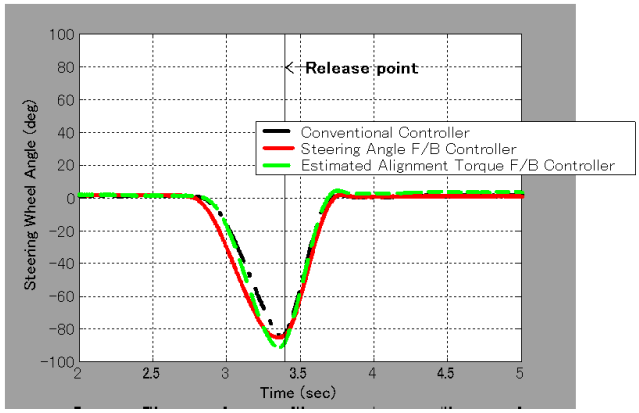


Figure 11 Test of releasing the steering wheel on a high μ road

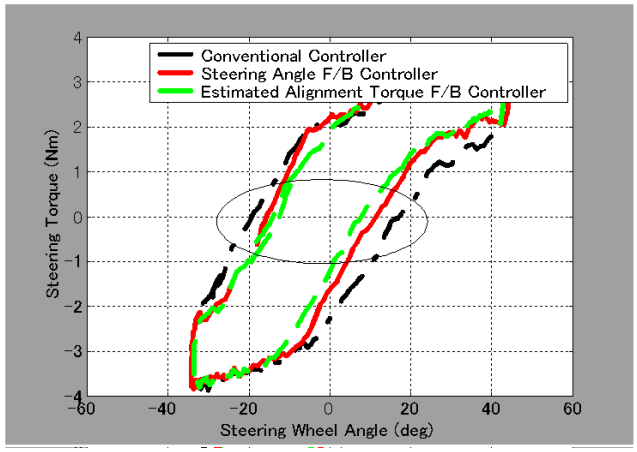


Figure 12 Test of sinusoidal steering on a high μ road

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CONCLUSIONS

A new control strategy to improve steering maneuverability was studied with the following results.

1. The proposed control algorithm improves steering wheel returnability on low μ roads.

2. Using the proposed controllers, the driver's sense of slipperiness is not lost on low μ roads.

3. Using the estimated alignment torque feedback controller, no supplemental sensors like steering angle sensors or motor angle sensors are required.

4. Using both of the two proposed controllers, no harmful subjective effects occur on high μ roads.

ACKNOWLEDGMENTS

We would like to thank the members of Himeji Works, Mitsubishi Electric Corp for their kind assistance in developing the new controller.

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NOMENCLATURE

T_{sens}: Measured steering torque

I_{mtr}: Motor current

V_t: Terminal voltage of motor

θ_{hdl}: Steering wheel angle

θ_s: Column axis angle

J: Moment of inertia of motor (at column axis)

T_{hdl}: Driver's steering torque

T_{tran}: Amount of reaction torque from below the column (rack & pinion, tire, etc. i.e. RTSS)

K_t: Torque constant of the motor

T_{f_mtr}: Frictional loss of the motor

T_{f_rp}: Frictional loss of the rack & pinion

G_{gear}: Reduction gear ratio from motor to column axis

T_{ASSIST}: Torque assisted by motor (at column axis)

T_{align}: Reduction torque of the tires (at column axis)